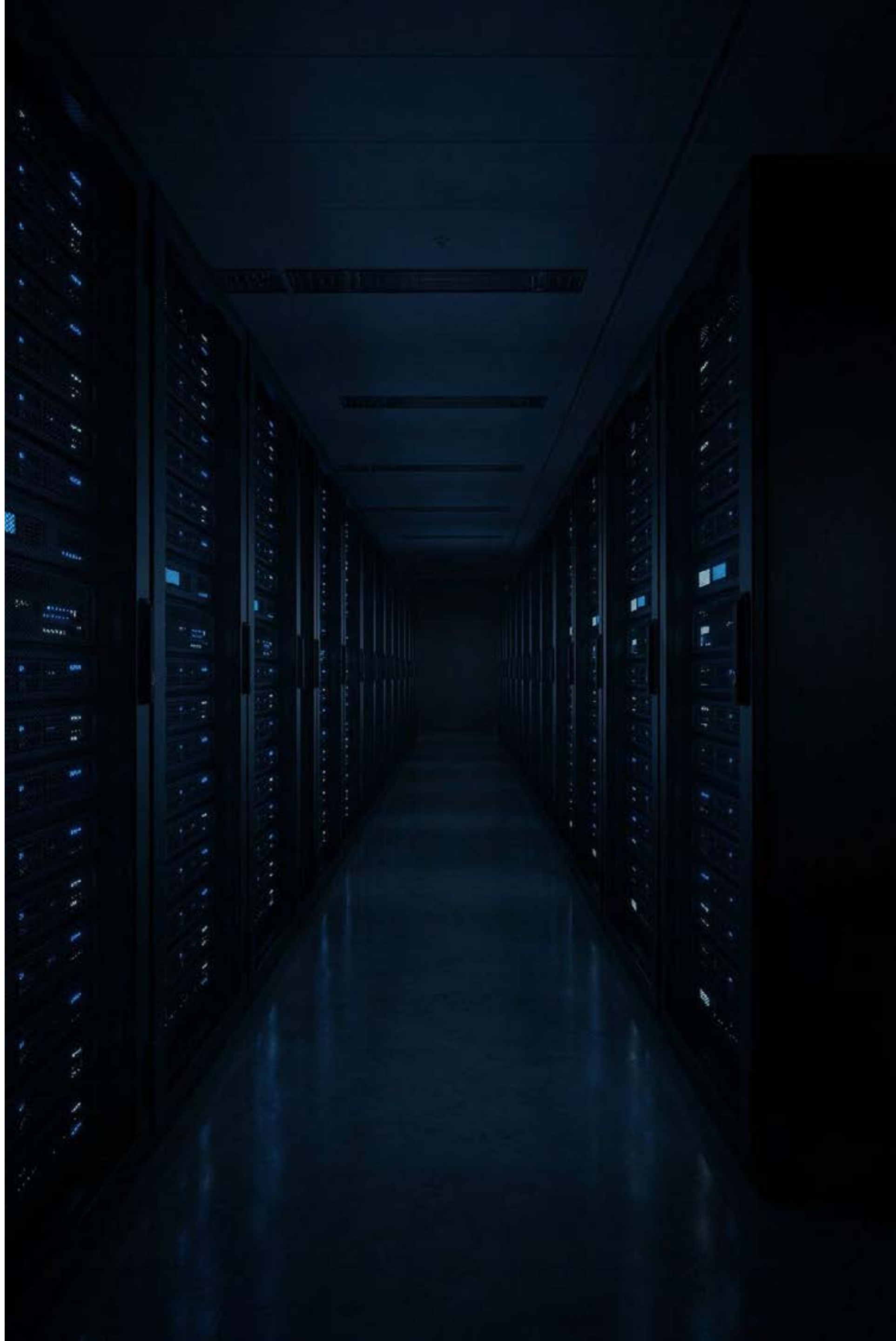
From Benchmarks to Business Impact: Evaluating Generative AI in Production

Conf42 Observability 2026



Presented By

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- 16+ years of experience in Generative AI, MLOps, and scalable backend systems
- Currently at **Adobe**, leading GenAI capabilities for Acrobat AI Assistant — evaluation pipelines, LLM orchestration, and AI infrastructure on AWS Bedrock & Azure
- Built observability frameworks (Grafana, New Relic) and drove key product growth initiatives
- Prior experience at **Informatica, CA Technologies, Oracle, and CSC** — identity & access management, SSO/SAML, cloud integration, and enterprise data systems
- **M.Tech** in Computer Science, VIT | Published researcher
- Expertise: Java, Python, Cloud Platforms, AI/ML Frameworks, DevOps & Observability

Why Traditional Benchmarks Fall Short

THE CORE PROBLEM

A model that performs well on benchmarks can still fail unpredictably in production. Benchmarks compare model capabilities — they rarely capture the complexity of real-world AI systems.

Traditional Benchmarks

- Static, curated datasets
- Single-turn interactions
- Controlled, predictable inputs
- Academic leaderboard metrics
- Narrow capability snapshots

Production Reality

- Dynamic, unpredictable user inputs
- Multi-turn, context-dependent interactions
- Business-specific objectives
- End-to-end system dependencies
- Adversarial and unexpected inputs

From Model Evaluation to System Evaluation

Production evaluation must move beyond asking *"Is the model response good?"* toward a broader, more accountable question:

> **"Does the complete AI system consistently deliver reliable, safe, relevant, and valuable outcomes?"**

Measure

Define meaningful quality dimensions and measurable signals.

Validate

Combine automated evaluation with expert human judgment.

Monitor

Continuously detect degradation, drift, and emerging failures.

Govern

Establish traceability, accountability, and responsible AI controls.

What Should We Evaluate?

FOUR CORE EVALUATION DIMENSIONS

Factual Accuracy

Is the information correct and verifiable? Key signals: **hallucination rate**, citation accuracy, and ground-truth comparison.

Grounding

Is the response supported by authoritative context? Key signals: **retrieval relevance**, faithfulness, and source attribution.

Safety

Does the system behave within defined policy boundaries? Key signals: red-team testing, **policy compliance**, adversarial testing.

Task Completion

Did the system accomplish the user's objective? Key signals: **task success rate**, resolution rate, downstream action outcomes.

Make AI Quality Observable

Production evaluation requires measurable signals that can be monitored continuously metrics must reflect the application's risk profile and business objective.

Key Signal Types

Hallucination rate, response relevance, user feedback, and task success rate

Continuous Monitoring

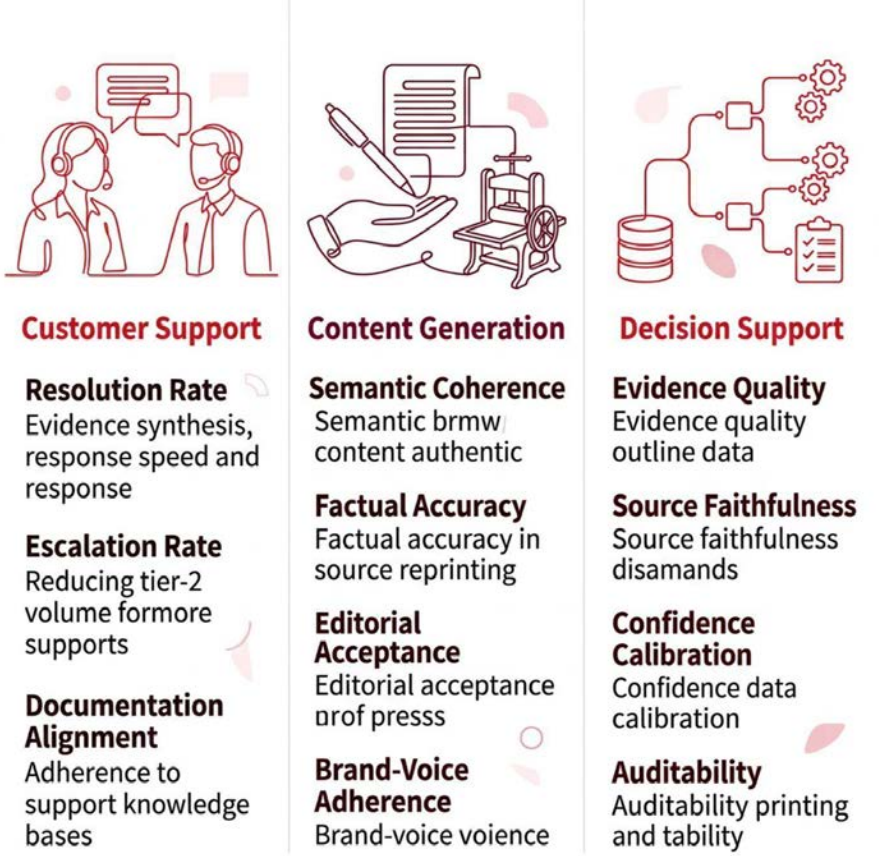
Metrics are not one-time snapshots they must be tracked across every production interaction

Guiding Principle

Every metric should connect directly to business objectives and user outcomes

Evaluation Must Be Domain-Specific

One evaluation framework does not fit every AI system. Each use case demands its own primary evaluation focus and quality signals.



Why This Matters

A customer support AI optimized for resolution rate is fundamentally different from a decision-support AI where evidence traceability and confidence calibration are paramount. Applying generic metrics to specialized domains produces misleading scores and false confidence.

Tailor your evaluation rubric to your domain then connect it to measurable business outcomes.

Automated Evaluation + Human-in-the-Loop

SCALE WITH AUTOMATION. VALIDATE WITH HUMANS.

Neither fully automated evaluation nor manual review alone is sufficient. The key is intelligent, risk-based sampling that routes the right interactions to human reviewers.

Automated Evaluation

LLM-as-a-judge, embedding similarity, classifier-based validation, rule-based checks, automated factuality and safety checks

Intelligent Sampling

Prioritize low-confidence, high-risk, policy-sensitive, and statistically significant edge cases for human review

Human Validation

Domain experts evaluate ambiguous, complex, and safety-critical scenarios preserving judgment where it matters most

Building a Continuous Quality Loop

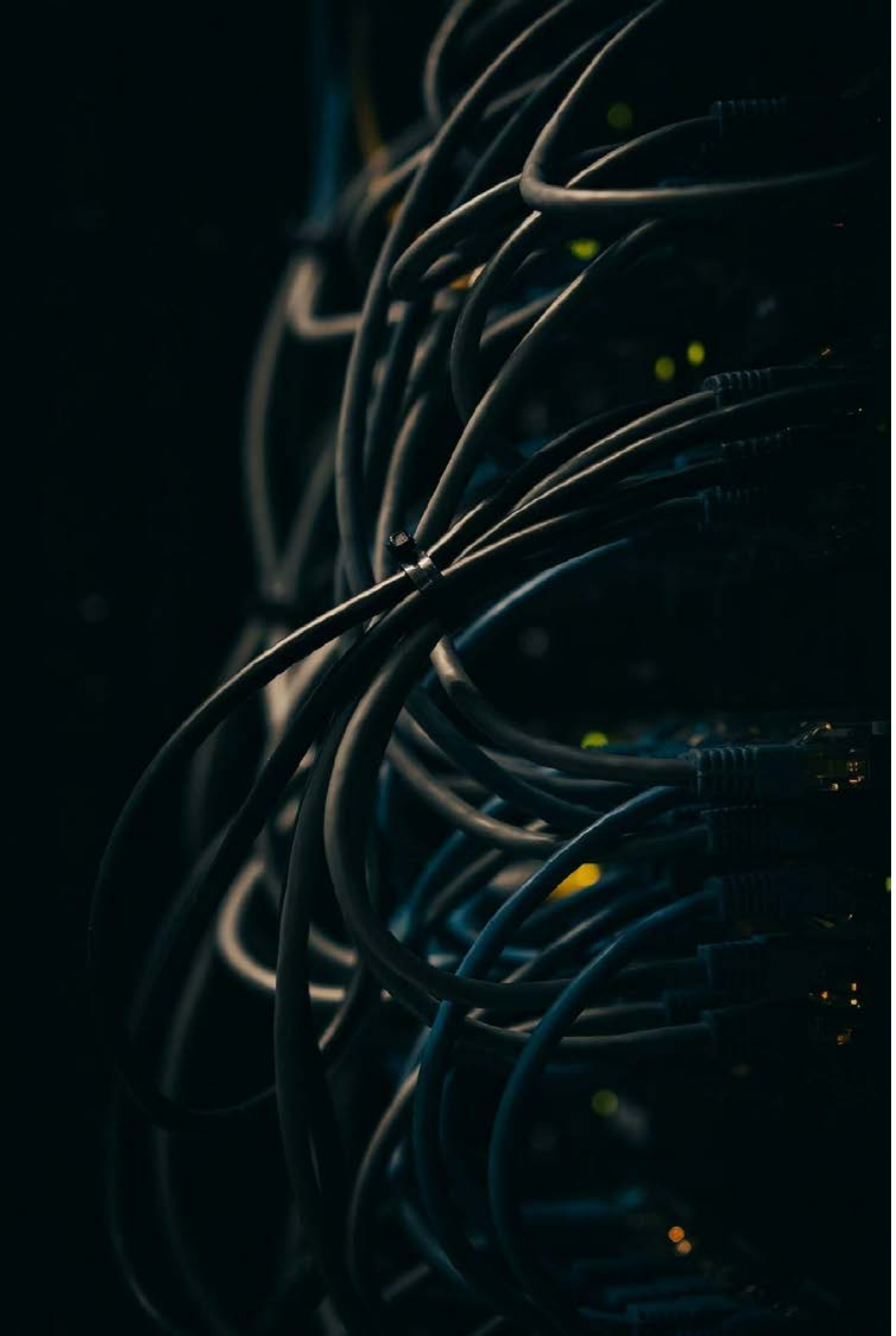
Human evaluation should not end with a score. It should feed back into the system continuously improving evaluation models, scoring rubrics, ground-truth datasets, and annotation guidelines.

From Gate to Learning System

Traditional evaluation is a one-time gate before release. A continuous quality loop transforms evaluation into a living system every production failure becomes an opportunity to improve both the AI and the evaluation framework that governs it.

- Evaluation models refined over time
- Ground-truth datasets kept current
- Judge prompts tuned to emerging failure modes

Production Interaction	**Automated Evaluation**
Risk-Based Sampling	**Human Validation**



Hallucination: Detect Before It Reaches Users

LAYERED HALLUCINATION DETECTION

Hallucination is one of the most significant production risks in generative AI. No single technique catches every failure a layered approach is required.

Entailment-Based Checking

Determine whether claims are logically supported by authoritative retrieved context.

Self-Consistency Sampling

Compare multiple generations to identify unstable or contradictory responses.

Retrieval Faithfulness

Measure whether response claims are directly supported by the retrieved evidence.

Inference-Time Controls

Apply confidence thresholds, constrained generation, and human escalation paths.

Goal: Prevent unreliable responses from reaching users not simply detect them afterward.

Robustness, Bias & Adversarial Inputs

PRODUCTION AI MUST SURVIVE UNEXPECTED INPUTS

Real users generate inputs that never appear in benchmark datasets. A production grade evaluation system must test against the full spectrum of adversarial and unexpected conditions.

Input Perturbation

Test against paraphrased prompts, noisy inputs, ambiguous requests, and adversarial variations to validate robustness across the input distribution.

Fairness Evaluation

Assess whether response quality or safety varies systematically across relevant demographic and contextual scenarios.

Red-Team Testing

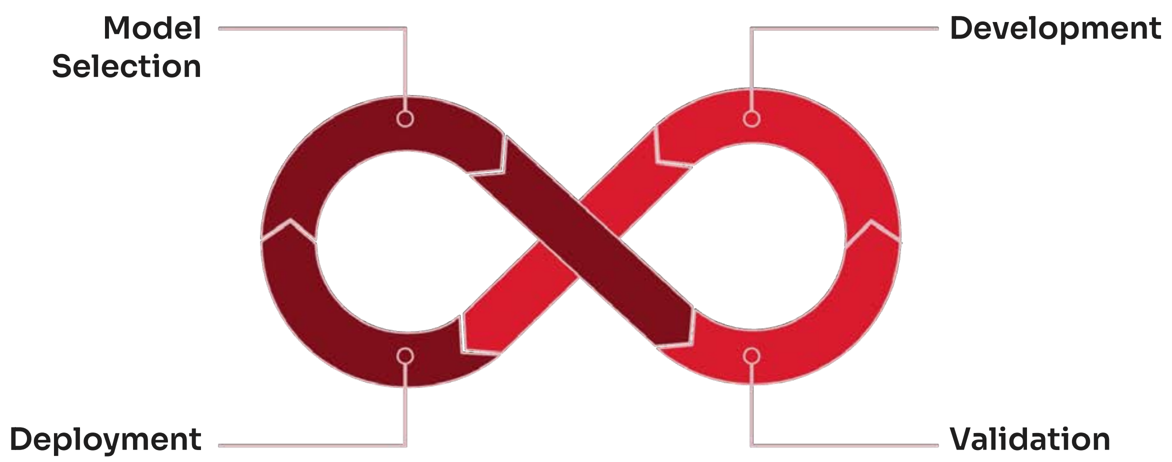
Proactively identify jailbreak vulnerabilities, prompt injection weaknesses, and policy edge cases before they reach production.

Distribution Shift Monitoring

Track how production inputs evolve over time and trigger re-evaluation when meaningful statistical shifts are detected.

Continuous Evaluation Across the AI Lifecycle

EVALUATION IS NOT A PRE-PRODUCTION GATE



What Continuous Evaluation Detects

- Model degradation over time
- New and unexpected failure modes
- Changing user behavior patterns
- Input distribution shifts
- Emerging safety and policy risks

AI systems evolve continuously and so must the evaluation systems that govern them. Treating evaluation as a pre-deployment gate creates blind spots in every stage that follows.

Production Evaluation & Observability Stack

MAKE AI BEHAVIOR VISIBLE

A mature evaluation architecture combines evaluation pipelines, observability tooling, interaction tracing, and governance controls into a single coherent system.

1

Real-Time Dashboards

Monitor hallucination rate, latency, refusal rates, user satisfaction, task success, and evaluation scores live and by segment.

2

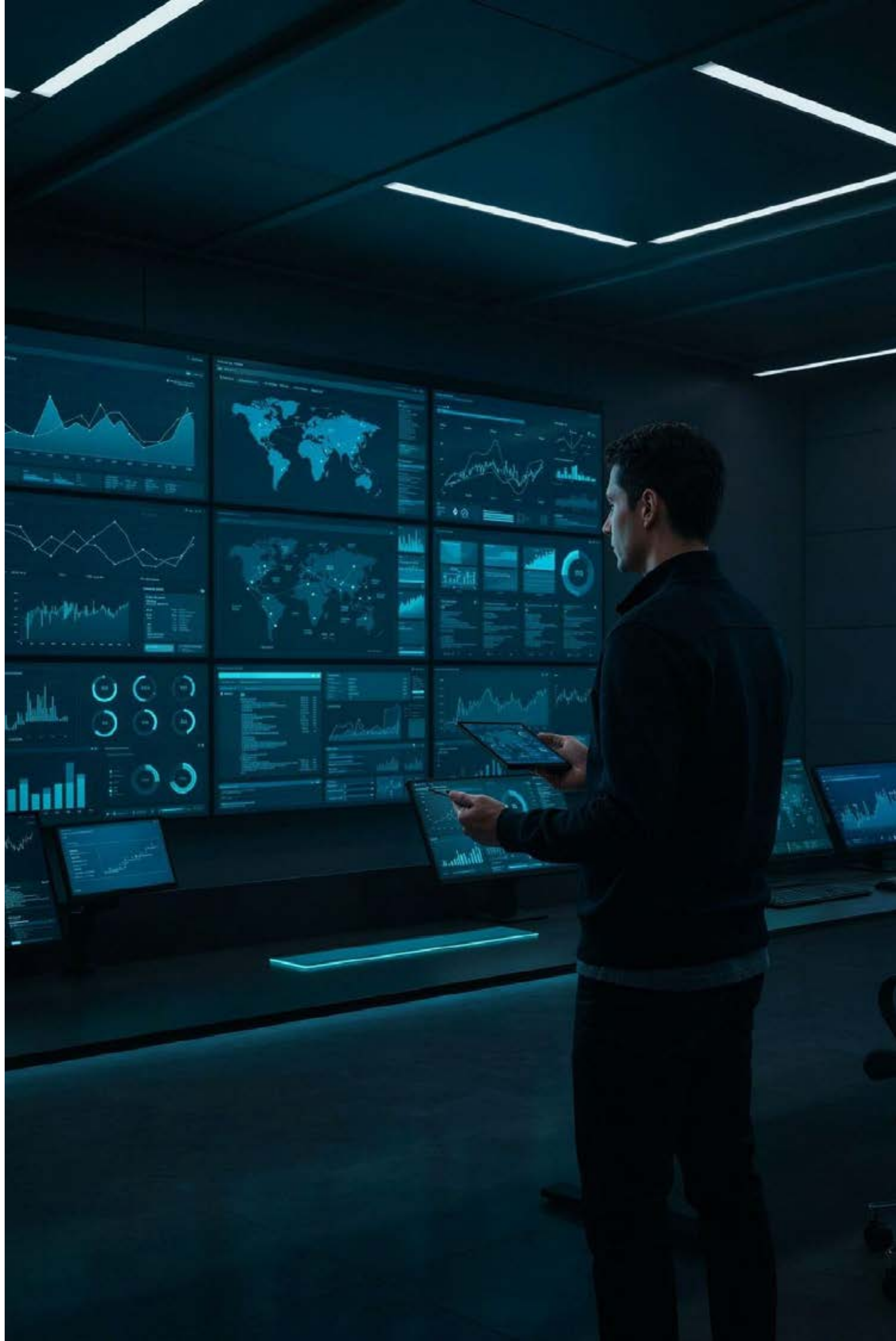
Segment Performance

Slice metrics by model version, prompt template, user segment, use case, and deployment environment to isolate failure sources.

3

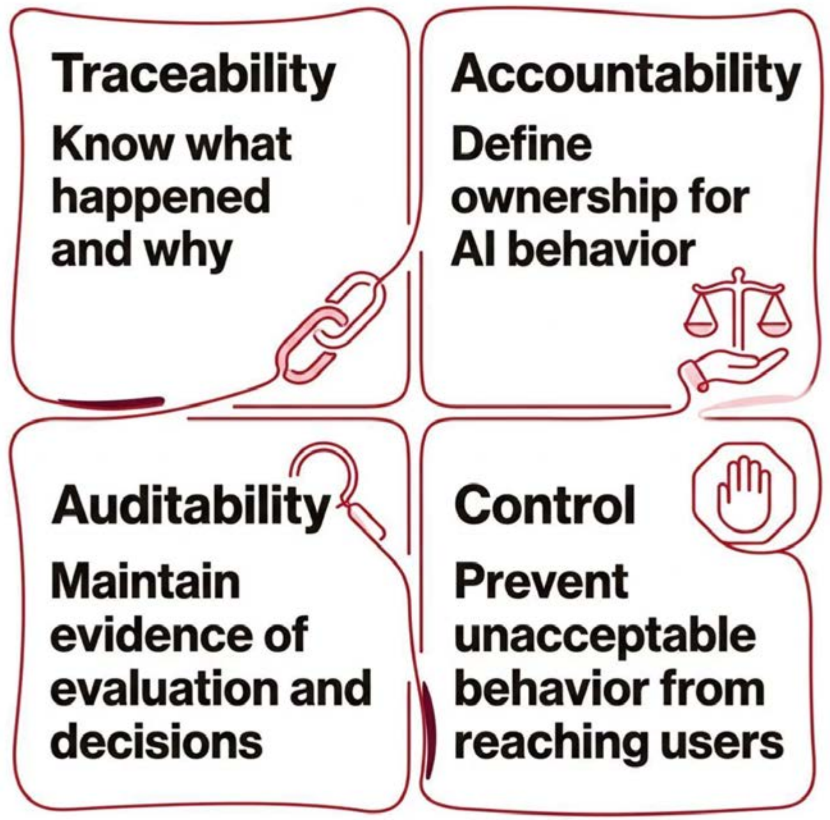
Full Interaction Tracing

Capture the complete chain: Input → Retrieved Context → Model Output → Evaluation Score → Metadata. Enables reproducibility and root-cause analysis.



Governance: Build Trust Into Engineering

Governance is not a compliance checkbox — it must be embedded directly into the AI delivery pipeline as a first-class engineering discipline.



Governance Capabilities

- Model versioning and evaluation snapshots
- Automated rollback triggers
- Human-escalation SLAs
- Audit logs and evaluation gates
- Compliance controls and documented methodologies

Governance built into engineering creates durable trust governance bolted on afterward creates audit theater.

Connect AI Metrics to Business Outcomes

TECHNICAL METRICS MATTER WHEN THEY EXPLAIN BUSINESS VALUE

Do not optimize AI metrics in isolation. Connect technical quality to measurable business outcomes creating a shared language across engineering, product, operations, and leadership.

Customer Support

Task Success → Higher Resolution Rate → Lower Escalation → **Lower Operational Cost**

Content Generation

AI Quality → Editorial Acceptance Rate → **Higher Team Productivity**

Decision Support

Evidence Quality → Better Recommendations → **Improved Downstream Decisions**

Key Takeaways: Building AI Systems You Can Trust

1 **Go Beyond Benchmarks**

Static scores do not guarantee production reliability. Evaluate the complete system.

2 **Measure What Matters**

Track factuality, grounding, safety, task success, and real user outcomes.

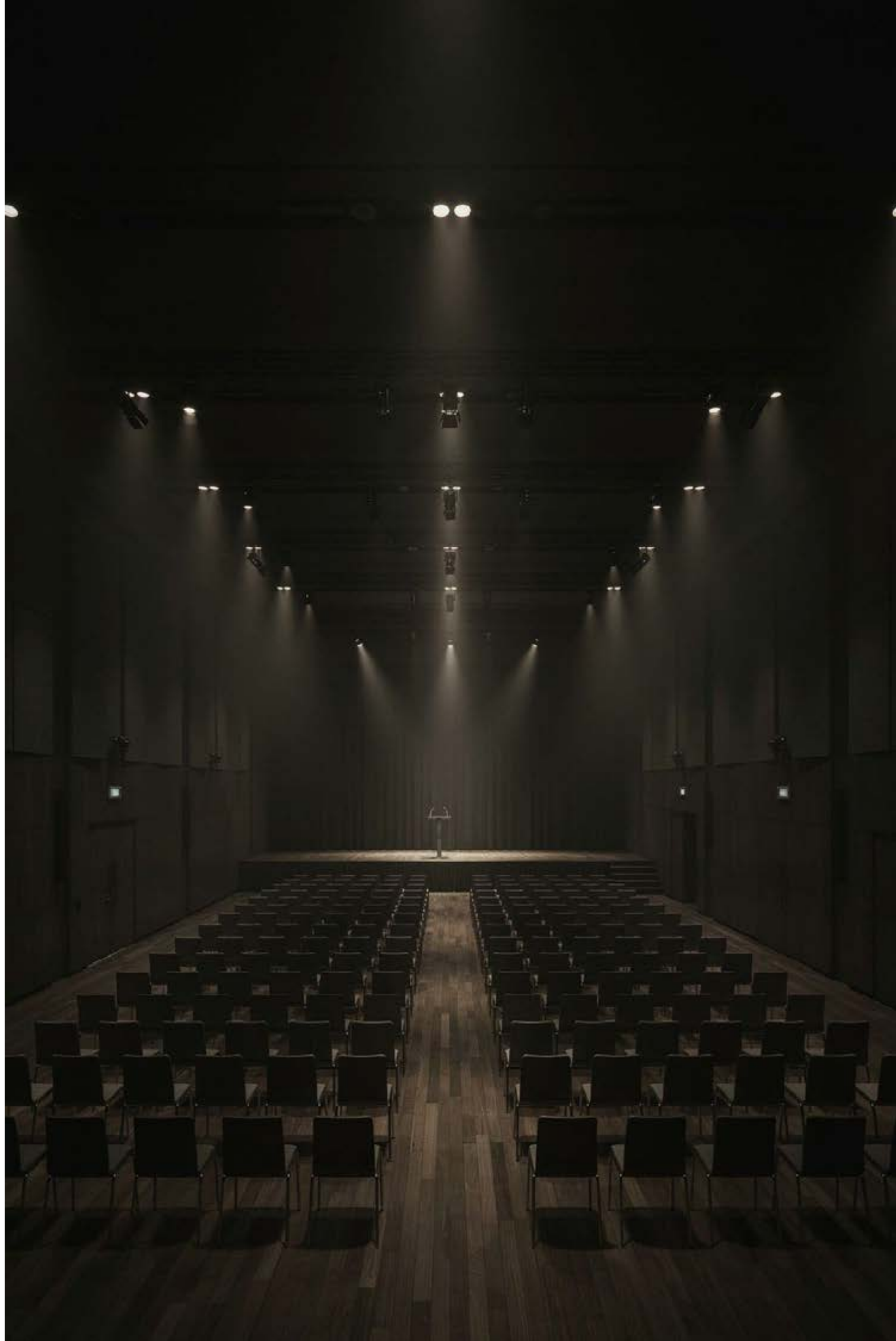
3 **Combine Automation + Human Judgment**

Automate scale preserve expert validation for complex and high-risk scenarios.

4 **Evaluate Continuously & Govern by Design**

Treat evaluation, monitoring, traceability, and accountability as core engineering disciplines not afterthoughts.

> The ultimate measure of AI is not model performance it is **reliable, governed, and measurable outcomes** in production.



Thank You

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